STA 3050: TIME SERIES ANALYSIS AND FORECASTING

Credits: 3 Hours
Prerequisite: STA 2030: Statistical Inference, STA 3010 Statistical Modelling

3.35.2 Purpose of the Course

The purpose of this course is to introduce a variety of statistical models for time series and cover the main methods for analyzing these models.

3.35.3 Expected Learning Outcomes of the Course

Upon successful completion of the course the learner will be able to:

1. Understand the basic concepts of time series analysis
2. Apply the concept of stationarity to the analysis of time series data
3. Run and interpret time-series models and regression models for time series
4. Use appropriate statistical software such as R, Python, SPSS or STATA to conduct the appropriate time series data analyses.

3.35.4 Course Content

Introduction to Time Series Analysis: definitions, time and frequency domain (spectral) methods and their applications; **Visualizing Time Series Data:** time series/ sequence plots, decomposing time series data; **Stationarity, Differencing, and Autocorrelation:** Stationary and non-stationary time series, Differencing to achieve stationarity, Autocorrelation and partial autocorrelation functions; **Time Series Forecasting Methods:** Moving average techniques and exponential smoothing methods (simple, Holt's Linear and Holt-Winters), Autoregressive (AR), Moving Average (MA), and ARMA models; **ARIMA Models:** Understanding autoregressive integrated moving average (ARIMA) models, Model identification, estimation, and diagnostic checking, Forecasting using ARIMA models

3.35.5 Instructional Methods

Micro-lectures, instructional videos, posting course notes, online discussion forums, written assignments, self-directed learning, reflective portfolios, and group work projects.

3.35.6 Instructional Materials and Equipment

Learning Management System, Internet and Web-based materials, microcomputer devices e.g. desktop, laptop, core course textbooks, articles, e-books, e-journals, and suggested readings.

3.35.7 Course Assessment

Attendance and participation	10%
Quizzes	10%
Group Work (Case Studies)	10%
Term Paper	20%
Mid-Trimester Exam	20%
Final Examination	30 %
Total	**100%**

3.35.8 Core Reading Materials for the Course

a). Textbooks

1. Chatfield (2016). *The Analysis of Time Series*, (7th ed.). CRC Press.
2. Robert H. Shumway & David S. Stoffer (2016). *Time Series Analysis and its Applications with R Examples* (3rd ed.). Springer.

b). Journals

1. Journal of Time Series Analysis
2. Computational Statistics & Data Analysis Journal

c). E-materials

1. *http://onlinelibrary.wiley.com/book/10.1002/9781118032978*
2. *https://www.stat.pitt.edu/stoffer/tsa4/*

3.35.9 Recommended Reference Materials

a). Textbooks

1. Rabe-Hesketh, S., & Skrondal, A. (2012). *Multilevel and Longitudinal Modeling Using STATA*, (3rd ed.).
2. Molenberghs, G. and Kenward, M. G. (2007). *Missing Data in Clinical Studies*. Chichester, UK:Wiley

b). Journals

1. Journal of American Statistical Association
2. International Journal of Forecasting
3. Econometrica

c). E-materials

1. *http://coin.wne.uw.edu.pl/pwojcik/tsa/pmwiki/pmwiki.php*
2. *https://www.youtube.com/playlist?list=PLJHszsWbB6hrkmmq57lX8BV-o-YIOFsiG*
3. *http://link.springer.com/chapter/10.1007%2F978-3-319-23446-5_9*